

# ***Quantum Entanglement as Geometric Obstruction***

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U.S. DEPARTMENT OF  
**ENERGY**



RESEARCH CORPORATION  
for SCIENCE ADVANCEMENT



# Motivations for this project

1. To establish a quantum information theory on an arbitrary parameter space  $X$ .
2. To discover new physics and verify experimentally.
3. To gain new math insights.

# Outline

## **Part 1: Math:** Entanglement as a new geometric obstruction

Main paper: [arXiv:2601.13764 \[math.AG\]](#),  
Lecture note [arXiv:2601.19111 \[quant-ph\]](#)

## **Part 2: Condensed Matter:** New observable beyond Berry phase

Yaron Oz [arXiv:2604.11596 \[cond-mat.mes-hall\]](#)



Main Math



Lecture Note



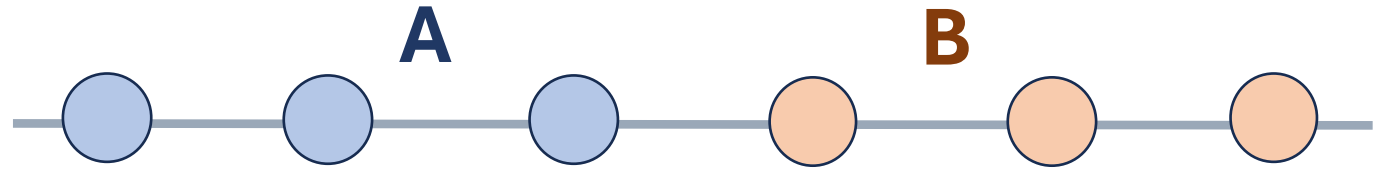
Condensed Mater



## PART 0

# Introduction: Quantum Entanglement

# Quantum Entanglement



$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

$\mathcal{H}_i$ : finite dim complex Hilbert space ( $\mathbb{C}^{\dim_i}$ )

Product state

$$|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle \quad \exists |\psi_i\rangle \in \mathcal{H}_i$$

**Classical: No quantum correlations**

Entangled state

$|\psi\rangle$  cannot be written as  $|\psi_A\rangle \otimes |\psi_B\rangle$   
with any  $\psi_A, \psi_B$

**Genuine Quantum**

# Very Important Observation

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**Entanglement of a state  $|\psi\rangle$  is a subsystem dependent notion**

**Pick up a state  $|\psi\rangle$  (this is always possible)**

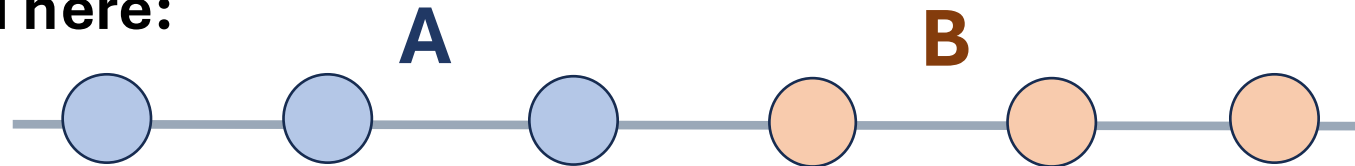


# Very Important Observation

Entanglement of a state  $|\psi\rangle$  is a subsystem dependent notion

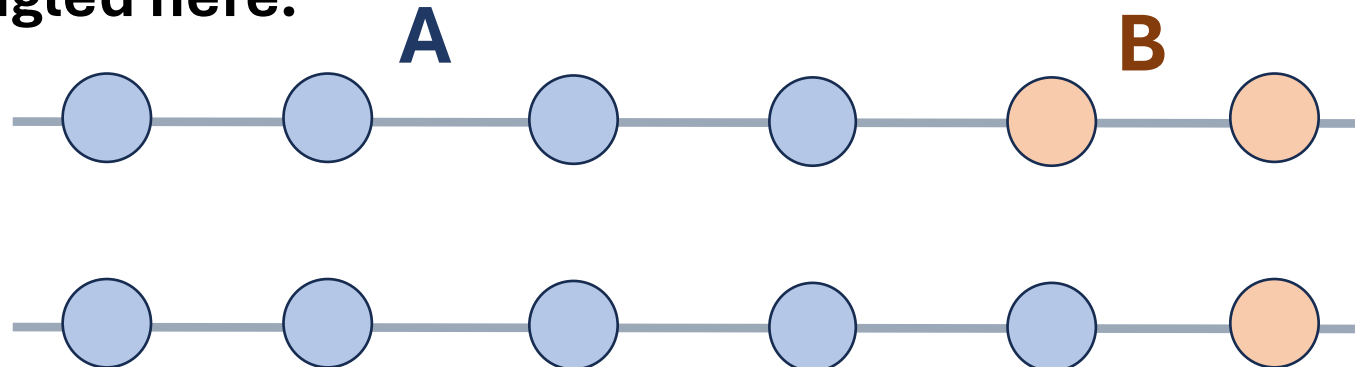
$|\psi\rangle$  can be entangled here:

$$|\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$



$|\psi\rangle$  may not be entangled here:

$$|\psi\rangle = |\varphi_A\rangle \otimes |\varphi_B\rangle$$



# Quantum Mathematics

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## Informal definition:

Conventional Math + **Entanglement**



Need information about *quantum correlations* within a space

## Common confusions:

Just having algebra ( $C^*$ / von Neumann/ Azumaya ...) does not immediately mean quantum.



# Quantum Mathematics

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1. Of course, conventional math is still powerful
2. But a way to look at objects is different.
3. Consider **quantum correlation** between objects



This requires an additional information: “**subsystem structure**”

# Take Away for a New Horizon

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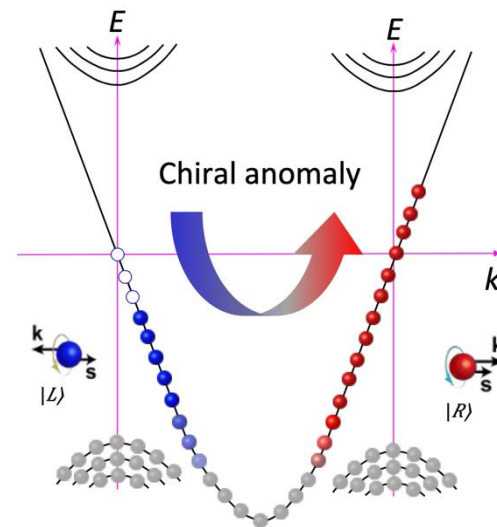
In other words, most of the subsystem independent quantities (eg. homology, homotopy invariants) are NOT very helpful to study QIS

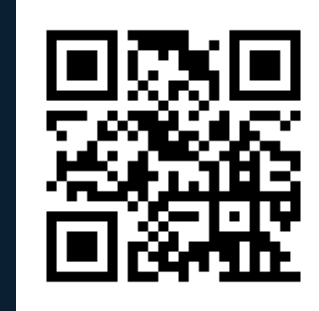
**They wash away quantum information** when making themselves subsystem independent.

# Take Away for a New Horizon

In other words, most of the subsystem independent quantities (eg. homology, homotopy invariants) are NOT very helpful to study QIS

Eg: AS index tells imbalance b/w #left & #right movers  
$$\nu_+ - \nu_- = \frac{1}{2\pi} \int_M \text{Tr } F$$
, but does not tell entanglement  
between them





## PART 1

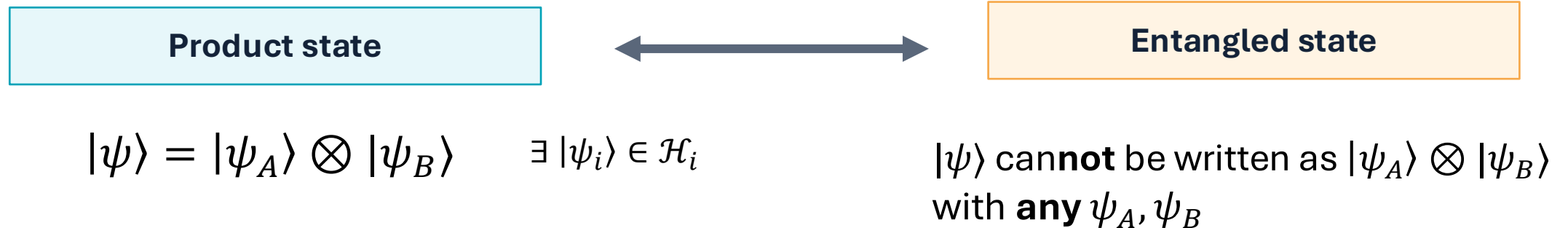
# Mathematical core

**Entanglement is local-to-global obstructions.**

# Several questions I had when I was a student...

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1. Why is it defined in such a convoluted & negative way?  
Obviously it is not straightforward to judge whether it is entangled or not:



2. How to look at this from geometry to understand it more intuitively?

# Observation: Geometry & Entanglement

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If  $|\psi\rangle$  is product

$|\psi\rangle$  can be **reconstructed** from **local information**



If  $|\psi\rangle$  is entangled

$|\psi\rangle$  cannot be reconstructed from local information



**Reconstruction is obstructed:** Local-to-global failure

# Observation: Geometry & Entanglement

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1. This idea is precisely what we learned for the characteristic classes (Chern, Euler, ...)
2. But characteristic classes do not depend on a choice of coverings  $\mathcal{U} = \{U_i\}$
3. How to close this conceptual gap? ← **Answered soon**

If  $|\psi\rangle$  is entangled

$|\psi\rangle$  cannot be reconstructed from local information



Reconstruction is obstructed: **Local-to-global failure**

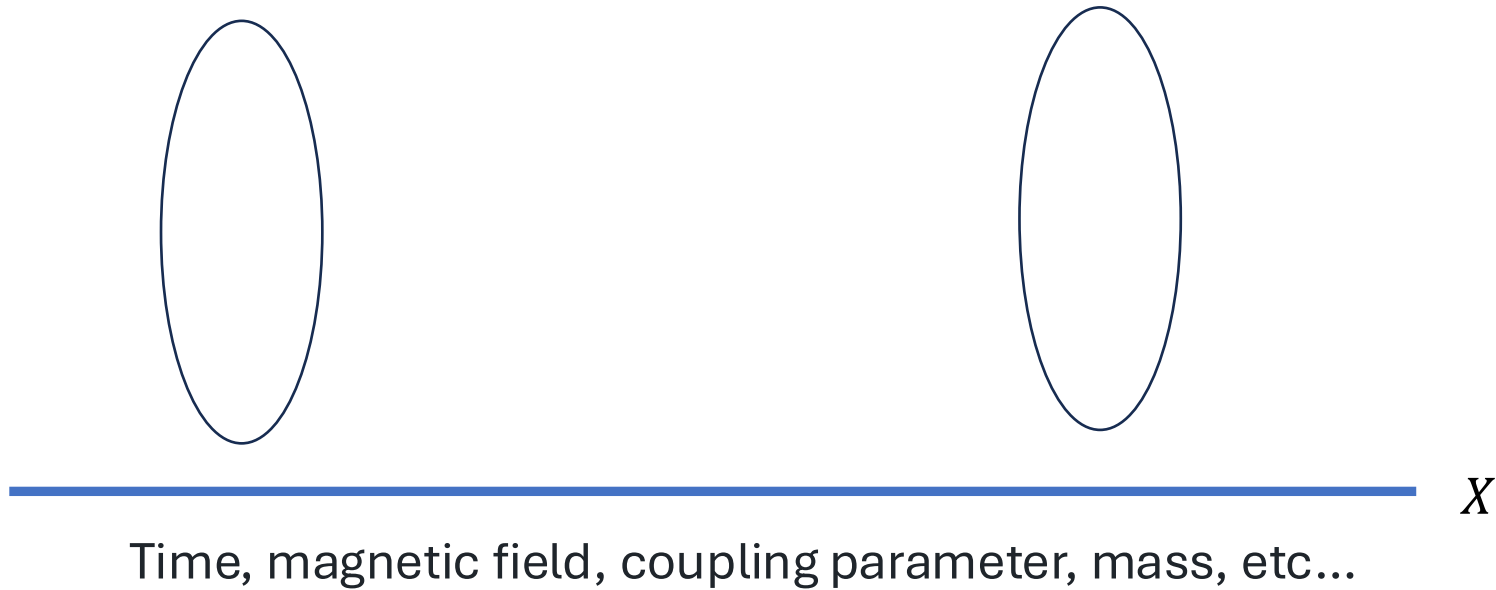
# Entanglement & Moduli Space

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$$\mathbb{P}(H_A) \times \mathbb{P}(H_B) \hookrightarrow \mathbb{P}(H_A \otimes H_B)$$

$$\psi = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$$

$$a(x), b(x), c(x), d(x), x \in X$$





# Entanglement & Moduli Space

---

$$\mathbb{P}(H_A) \times \mathbb{P}(H_B) \hookrightarrow \mathbb{P}(H_A \otimes H_B)$$

$$\psi = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle \quad M_\psi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

**Algebraic variety:**

$$\Sigma_{(2,2)} = \{\det M_\psi = ad - bc = 0\} \subset \mathbb{P}^3$$

Concurrence = 0  $\iff$  Product state

Geometrically, product states are “clean” objects; always belong to an algebraic variety.

Entangled states are “wild”; belong to a Zariski open set.

The same observation holds for general N qubits with k-partitioning.

# Entanglement & Severi-Brauer Scheme

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$$SB(A) = P \times^{PGL_n} \mathbb{P}^{n-1} \longrightarrow X$$



## Dictionary

$X$  : parameter space

$P \rightarrow X$  : principal  $PGL_n$ -torsor of projective coordinate changes

$A$  : Azumaya algebra: locally an  $n \times n$  matrix algebra

This is a well-known setup.

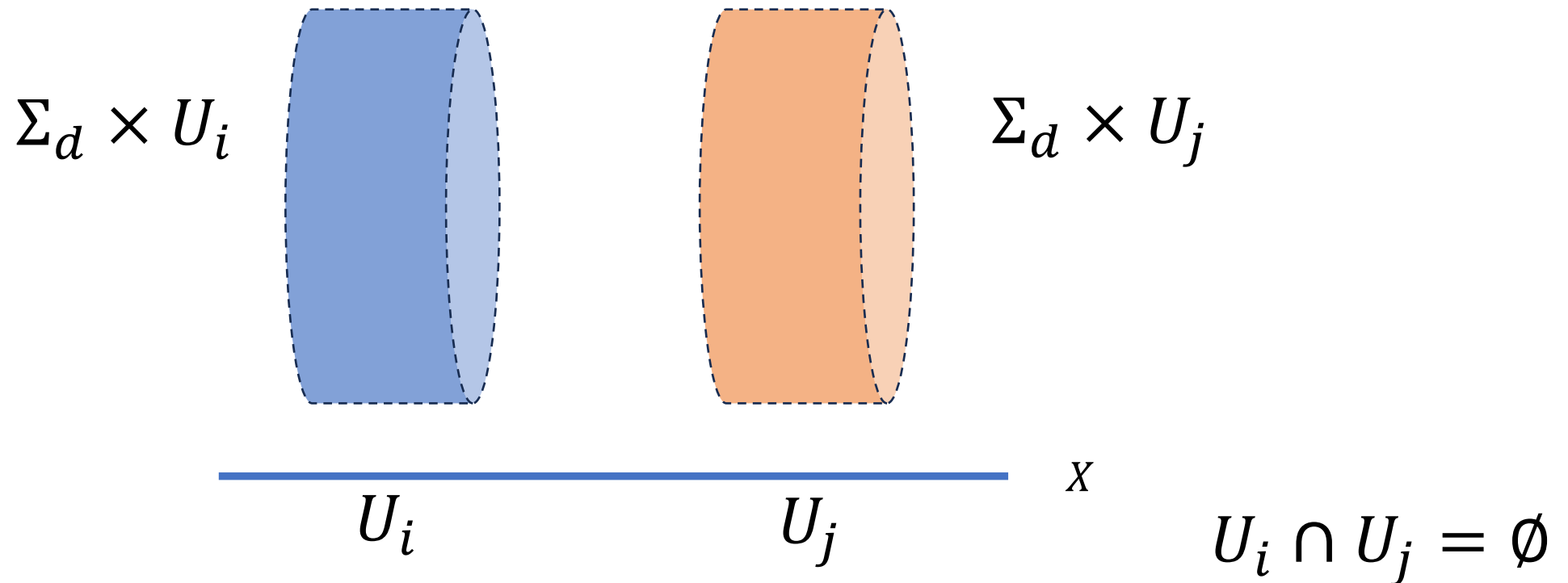
What is missing for entanglement?

To discuss entanglement, we need to specify a **subsystem**

$$\Sigma_d = \mathbb{P}^{d_1-1} \times \dots \times \mathbb{P}^{d_s-1} \hookrightarrow \mathbb{P}^{n-1}, \quad n = \prod_i d_i$$
$$G_d = \{g \in PGL_n \mid g(\Sigma_d) = \Sigma_d\}$$

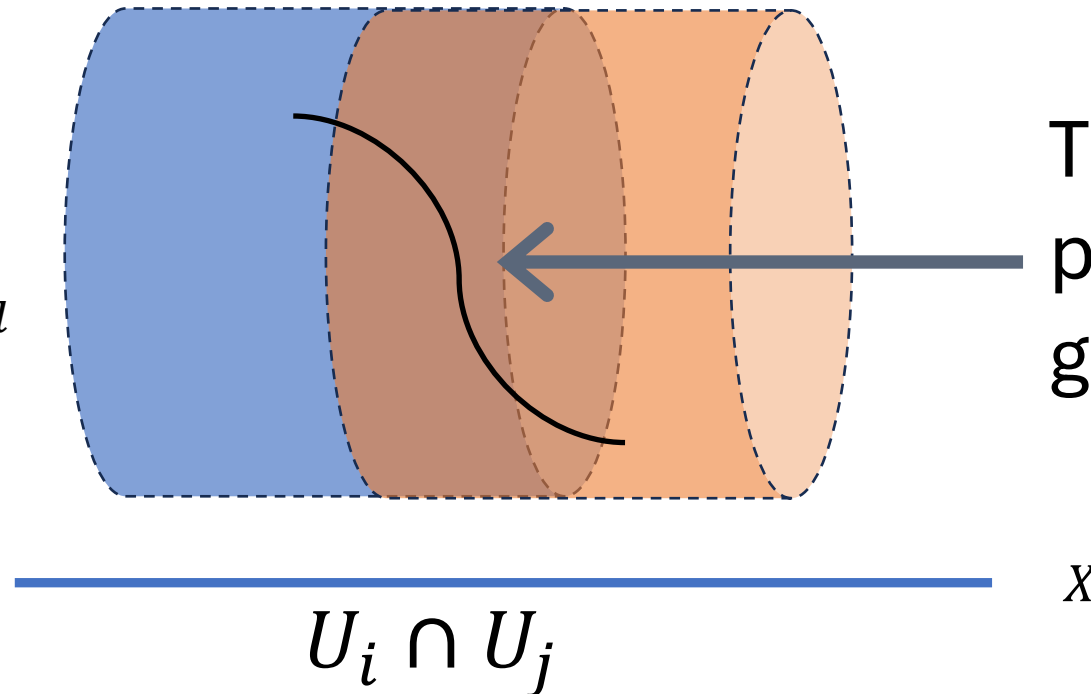
Chose a local trivialization  $SB(A)|_{U_i} \simeq \mathbb{P}^{n-1} \times U_i$

Embed a subsystem  $\Sigma_d = \mathbb{P}^{d_1-1} \times \dots \times \mathbb{P}^{d_s-1} \hookrightarrow \mathbb{P}^{n-1}, \quad n = \prod_i d_i$



However,  $\Sigma_d \times U_i$  and  $\Sigma_d \times U_j$  can be different on overlap  $U_i \cap U_j \neq \emptyset$

Gluing  $g \in PGL_n$  on  
 $U_i \cap U_j$  may not be in  $G_d$



This “twist” can  
produce entangling  
gluing

How to encode the information of subsystem into the SB-scheme

Main Theorem (KI: arXiv:2601.13764):

$$P/G_d \simeq \mathrm{Hilb}^{\Sigma_d}(SB(A)/X)$$

$P/G_d$  is  $\frac{\text{all possible projective frame choices}}{\text{local choices preserving product-state frame}}$

$\mathrm{Hilb}^{\Sigma_d}(SB(A)/X)$  is the moduli space of all product states with respect to  $\Sigma_d$

To have a globally consistent subsystem structure,  
there must be a global section  $X \rightarrow P/G_d$

**This is why entanglement is obstruction.**

**This is also true to N qubits with k-block partition.**

$$\underbrace{H^1(X, G_d)}_{\text{subsystem structure}} \longrightarrow \underbrace{H^1(X, PGL_n)}_{\text{projective family}} \xrightarrow{\delta} \underbrace{Br(X)}_{\text{Brauer class}}$$

**Information about quantum entanglement is encoded in  $G_d$ -valued 1-cocycle data**

Theorem (KI: arXiv:2601.13764 )

**Subsystem obstruction may survive even when Brauer class is trivial.  
(New obstruction, that I call entanglement obstruction)**



# Entanglement Obstruction

KI: arXiv:2601.13764

$$H^1(X, G_d) \longrightarrow H^1(X, PGL_n) \xrightarrow{\delta} Br(X)$$

subsystem structure

projective family

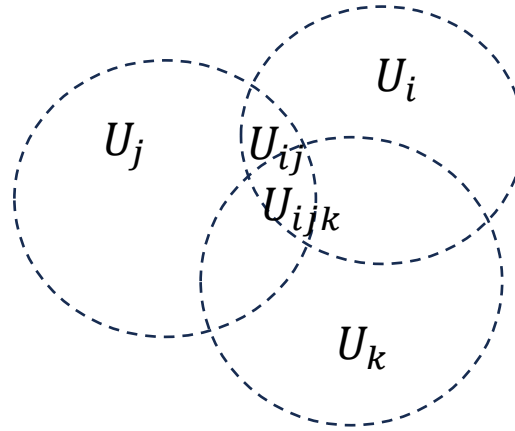
Brauer class

$$G_d \subset PGL_n$$

$G_d$ -valued 1-cocycle data

$$s_j = a_{ij}s_i \text{ (section)} \quad a_{ij} \in G_d(U_{ij})$$

$$a_{ij}a_{jk}a_{ki} = 1 \text{ on } U_{ijk}$$



Data for  $H^1(X, G_d)$

$$s'_i = s_i b_i^{-1} \quad b_i \in G_d(U_i)$$

$$a'_{ij} = b_i a_{ij} b_j^{-1}$$



Send this to  $PGL_n$  by inclusion

$$[c_{ijk}] = \beta \in Br(X)$$

Theorem (KI: arXiv:2601.13764 )

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# Entanglement Obstruction

KI: arXiv:2601.13764

$$H^1(X, G_d) \longrightarrow H^1(X, PGL_n) \xrightarrow{\delta} Br(X)$$

subsystem structure

projective family

Brauer class

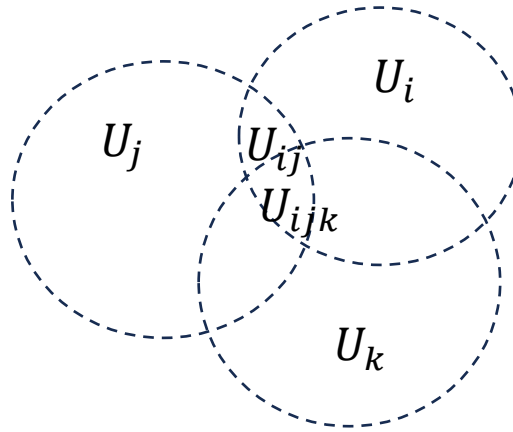
Interpretation

$$G_d \subset PGL_n$$

$G_d$ -valued 1-cocycle data

$$s_j = a_{ij}s_i \text{ (section)} \quad a_{ij} \in G_d(U_{ij})$$

$$a_{ij}a_{jk}a_{ki} = 1 \text{ on } U_{ijk}$$



$U_i$  is an open set in  $X$  (e.g. a time-segment)

$U_{ij}$  may yield entanglement (e.g. phase transition from  $U_i$  to  $U_j$ )

However, **entanglement really depends on  $\mathcal{U} = \{U_i\}$ , thus **not visible** in  $H^1$  & Br, which are ***independent*** of  $\mathcal{U}$**

Theorem (KI: arXiv:2601.13764 )

**Subsystem obstruction may survive even when Brauer class is trivial.  
(New obstruction, that I call entanglement obstruction)**

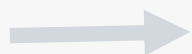
# Observation: Geometry & Entanglement

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1. This idea is precisely what we learned for the characteristic classes (Chern, Euler, ...)
2. But characteristic classes do not depend on a choice of coverings  $\mathcal{U} = \{U_i\}$
3. How to close this conceptual gap?  **It is a new obstruction invisible to conventional characteristic classes**

If  $|\psi\rangle$  is entangled

$|\psi\rangle$  cannot be reconstructed from local information



Reconstruction is obstructed: **Local-to-global failure**

## PART 2



# Validation with materials

**Entangling holonomies vs Berry/Chern/Wilson diagnostics**

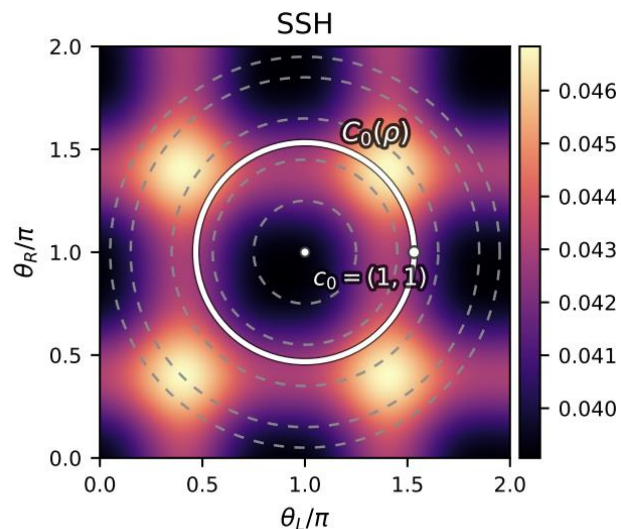
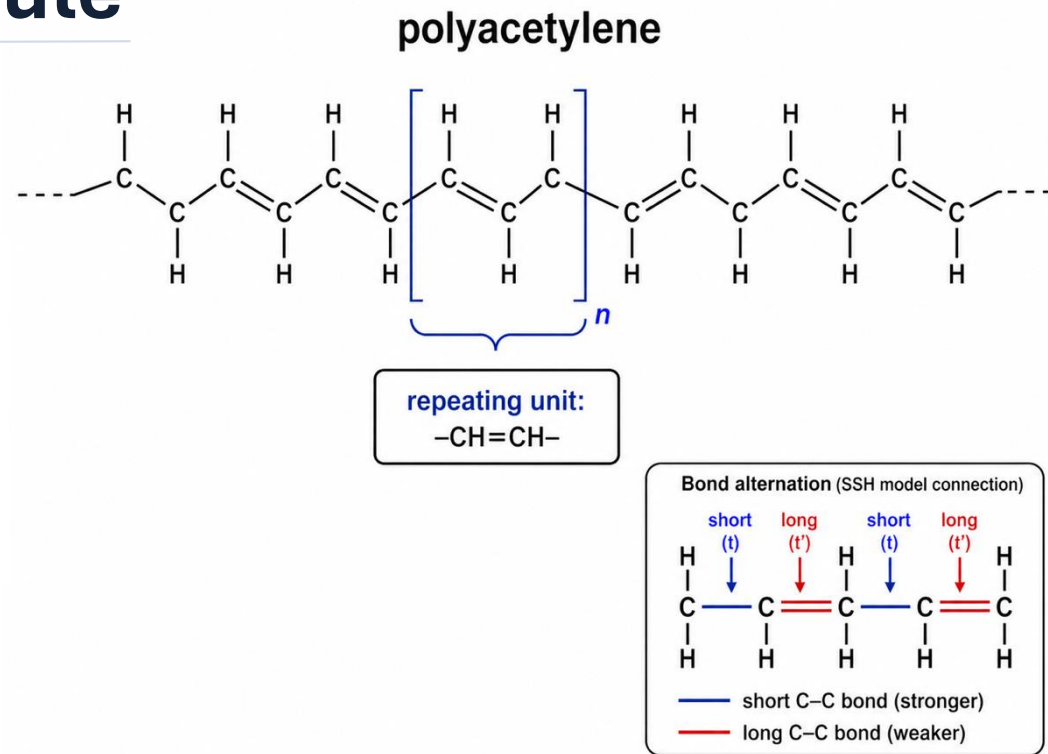
# Entanglement Obstruction in Molecule

Consider a molecular model (SSH-model)

$$H_{\text{SSH}} = \sum_{n=1}^N (c_{nA}^\dagger T_1 c_{nB} + \text{h.c.}) + \sum_{n=1}^{N-1} (c_{nB}^\dagger T_2 c_{n+1,A} + \text{h.c.}) \\ + B c_{1A}^\dagger (\cos \theta_L \sigma_x + \sin \theta_L \sigma_y) c_{1A} + B c_{NB}^\dagger (\cos \theta_R \sigma_x + \sin \theta_R \sigma_y) c_{NB},$$

with

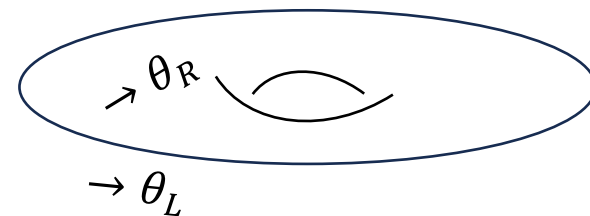
$$T_1 = t_1 I + i\lambda_1 \sigma_y, \quad T_2 = t_2 I + i\lambda_2 \sigma_x.$$



Entangling holonomy

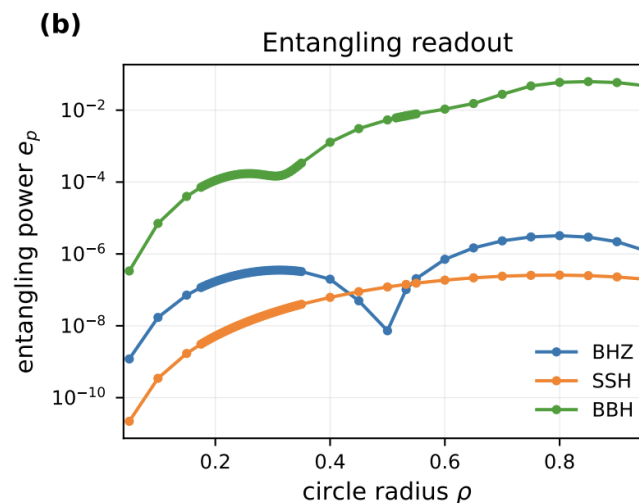
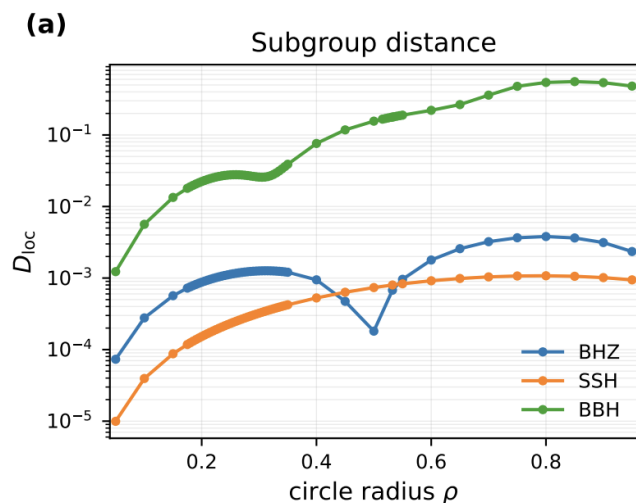
$$D_{\text{loc}}(C) = \min_{A, B \in U(2)} \|U(C) - A \otimes B\|_F$$

Low energy modes of polyacetylene :  
Twisted family of  $CP^3$  over  $T^2$

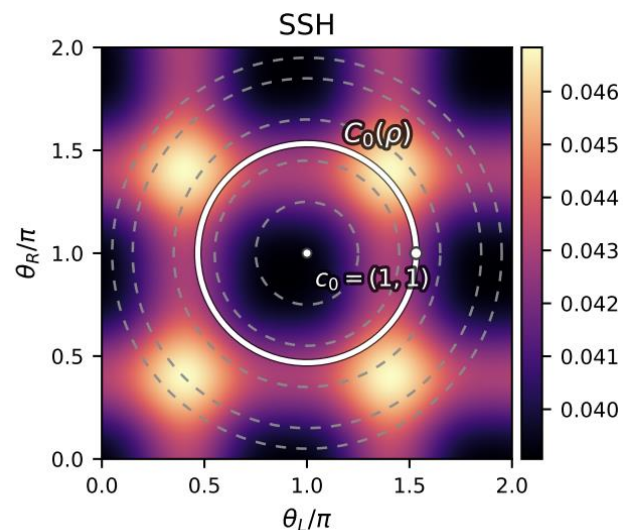


# Entanglement Obstruction in Molecule

KI & Oz: arXiv:2604.11596



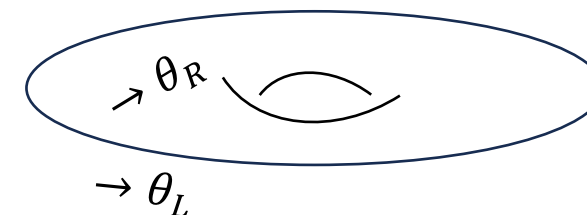
**Not detectable by homotopy /  
Brauer class / Chern class**



**Entangling holonomy**

$$D_{\text{loc}}(C) = \min_{A, B \in U(2)} \|U(C) - A \otimes B\|_F$$

Low energy modes of polyacetylene :  
Twisted family of  $CP^3$  over  $T^2$



# Bernevig-Hughes-Zhang (BHZ) model

KI & Oz: arXiv:2604.11596

$$H_{\text{BHZ}}(k_x, \theta_T, \theta_B) = \sum_{y=1}^{L_y} c_y^\dagger h_0(k_x) c_y + \sum_{y=1}^{L_y-1} (c_y^\dagger T_y c_{y+1} + \text{h.c.}) + c_1^\dagger V_T(\theta_T) c_1 + c_{L_y}^\dagger V_B(\theta_B) c_{L_y}, \quad (4.1)$$

with

$$h_0(k_x) = (M - 4B + 2B \cos k_x) \tau_z \otimes \sigma_0 + A \sin k_x \tau_x \otimes \sigma_z + \lambda_R \sin k_x \tau_x \otimes \sigma_y, \quad (4.2)$$

$$T_y = B \tau_z \otimes \sigma_0 - \frac{iA}{2} \tau_y \otimes \sigma_0 + \frac{i\lambda_R}{2} \tau_x \otimes \sigma_x, \quad (4.3)$$

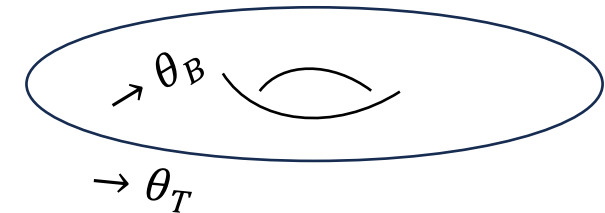
$$V_T(\theta_T) = h_e \tau_0 \otimes (\cos \theta_T \sigma_x + \sin \theta_T \sigma_y), \quad (4.4)$$

$$V_B(\theta_B) = h_e \tau_0 \otimes (\cos \theta_B \sigma_x + \sin \theta_B \sigma_y). \quad (4.5)$$

It is basically a model with “space(y)  $\otimes$  orbital ( $\tau$ )  $\otimes$  spin( $\sigma$ )”

Low energy modes of polyacetylene :  
Twisted family of  $CP^3$  over  $T^2$

$\theta_{B/T}$  are the directions of Zeeman fields



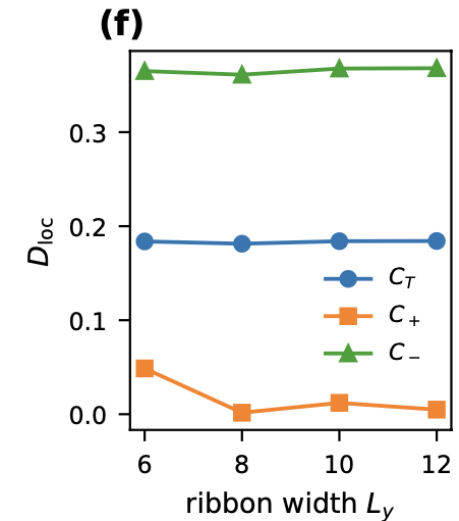
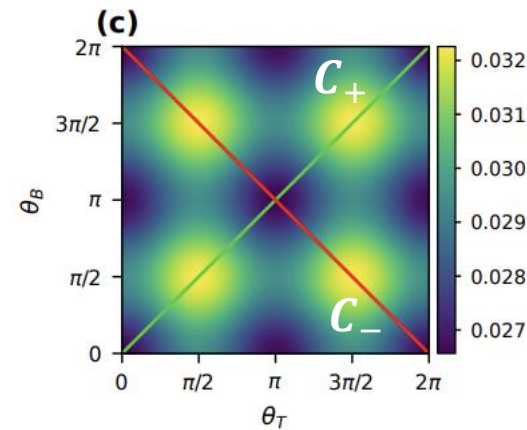
# Bernevig-Hughes-Zhang (BHZ) model

KI & Oz: arXiv:2604.11596

$$U(C_+) \approx I \otimes \exp(-i\phi Z) : \text{local gate}$$

$$U(C_-) \approx \exp(-i\phi Z \otimes Z) : \text{entangling gate}$$

$C_{\pm}$  yield completely different entangling holonomies



[Derivation] Suppose a holonomy is diagonal

$$U(C) \approx \text{diag}(e^{-i\alpha_{T+}}, e^{-i\alpha_{T-}}, e^{-i\alpha_{B+}}, e^{-i\alpha_{B-}})$$

Generator:  $aI \otimes I + bI \otimes Z + cZ \otimes I + dZ \otimes Z \longrightarrow$

$$\begin{aligned} \alpha_{T\pm} &= a + b \pm c \pm d, \\ \alpha_{B\pm} &= a - b \pm c \mp d \end{aligned}$$

Define  $\Delta := \alpha_{T+} - \alpha_{T-} - \alpha_{B+} + \alpha_{B-} = 4d$

$$\Delta = 0 \Leftrightarrow \text{Local (no interaction between edge and helical)}$$

$$\Delta \neq 0 \Leftrightarrow \text{Entangling}$$

Basis:  $|T, \pm\rangle, |B, \pm\rangle$

Top (T) / bottom (B) edges ,  
 $\pm$  helical spins



**Very very important: This cannot be detected by Berry / Chern number**

The non-Abelian Berry connection is  $A = A_T d\theta_T + A_B d\theta_B$

The connections along  $C_{\pm}$  are:  $A_{\pm} = A_T d\theta_T \pm A_B d\theta_B$

Then the generators of T/ B modes are  $K_{T/B} \sim \phi P_{T/B} \otimes Z$  with the projectors  $P_T = \frac{I+Z}{2}$  and  $P_B = \frac{I-Z}{2}$

$K_+ = K_T + K_B \sim \phi I \otimes Z$  (local operation)  $\longrightarrow$  The usual non-Abelian Berry connection

$K_- = K_T - K_B \sim \phi Z \otimes Z$  (entangling gate)  $\longrightarrow$  **This yields the entangling holonomy**

# Chern class vs Entangling Holonomies

| bundle               | quartet $\text{Ch}_1$ | block A $\text{Ch}_1$ | block B $\text{Ch}_1$ |
|----------------------|-----------------------|-----------------------|-----------------------|
| BHZ quartet / blocks | $\approx 0$           | $\approx 0$           | $\approx 0$           |
| SSH quartet / blocks | $\approx 0$           | $\approx 0$           | $\approx 0$           |
| BBH quartet / blocks | $\approx 0$           | $\approx 0$           | $\approx 0$           |

| model | loop              | $\arg \det U(\mathcal{C})$ | $\max_j  \phi_j $ | $D_{\text{loc}}(\mathcal{C})$ | $e_p(\mathcal{C})$ |
|-------|-------------------|----------------------------|-------------------|-------------------------------|--------------------|
| BHZ   | $C_T$             | $\approx 0$                | 0.18              | 0.18                          | 0.007              |
| BHZ   | $C_+$             | $\approx 0$                | 0.18              | 0.01                          | $< 0.001$          |
| BHZ   | $C_-$             | $\approx 0$                | 0.18              | 0.37                          | 0.03               |
| SSH   | $C_L$             | $\approx 0$                | 0.20              | 0.20                          | 0.009              |
| SSH   | $C_{\text{anti}}$ | $\approx 0$                | 0.20              | 0.38                          | 0.03               |
| BBH   | $C_y$             | $\approx 0$                | 0.83              | 0.008                         | $< 0.001$          |
| BBH   | $C_{\text{diag}}$ | $\approx 0$                | 1.26              | 0.15                          | 0.005              |

**Not detectable by homotopy /  
Brauer class / Chern class**

# Summary of Results & Significances

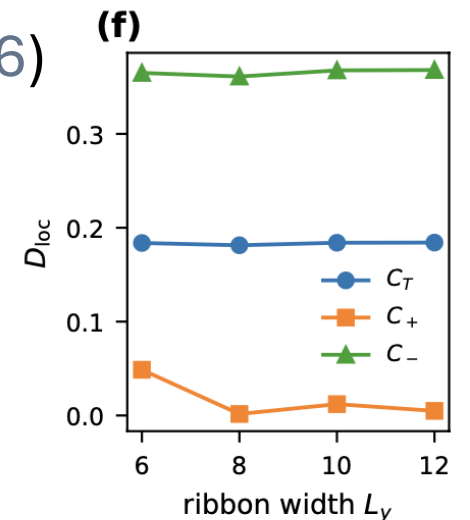
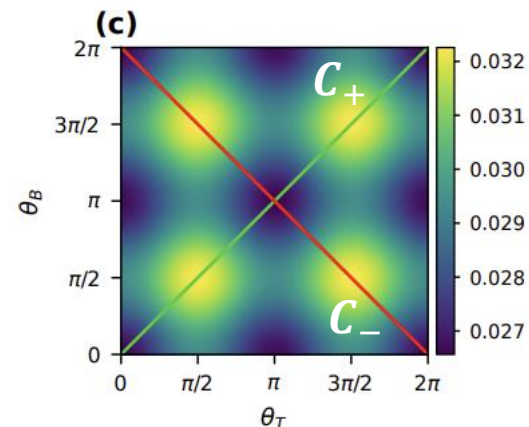
I established the math concept “Entanglement is Obstruction”

Theorem (Kl: arXiv:2601.13764):

$$P/G_d \simeq \text{Hilb}^{\Sigma_d}(SB(A)/X)$$

and proved such an obstruction may exist even when Brauer class is 0

With Y. Oz, we validated this with well-known condensed matter models: **loop-dependent entangling holonomies** (arXiv:2604.11596)



# Summary of Results & Significances

I established the math concept “Entanglement is Obstruction”

Theorem (Kl: arXiv:2601.13764):

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arXiv > math > arXiv:2601.13764

Mathematics > Algebraic Geometry

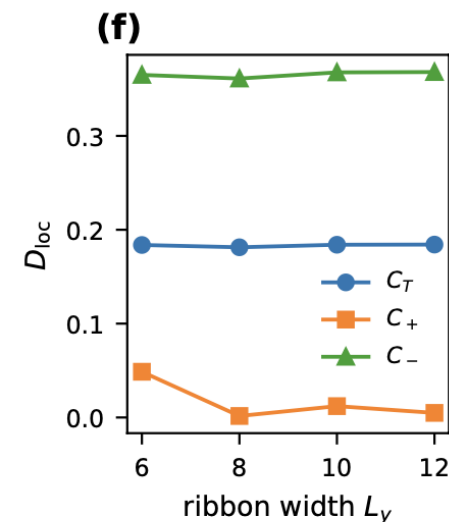
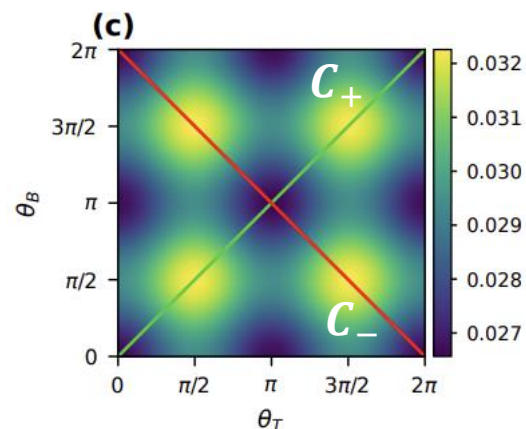
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[Submitted on 20 Jan 2026 (v1), last revised 26 Jan 2026 (this version, v2)]

## Quantum Entanglement Geometry on Severi–Brauer Schemes: Subsystem Reductions of Azumaya Algebras

Kazuki Ikeda

$C_-$  corresponds to subsystem reduction failure



# Summary of Results & Significances

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In a similar way, one can also construct the twisted geometry of

- Quantum channels/ Choi /Krause ops:  $SB(A_{in}^{op} \otimes A_{out})$
- Entanglement witnesses

This gives the most general identification of entanglement in any N-qubit system with any k-block partitioning, over arbitrary parameter space  $X$ .

# More directions

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## Extend the work to

- larger entangling holonomies
- mixed states / non-Hermitian
- infinite dim algebra ( $C^*$ / von Neumann type III )

Easy



Challenging

**We are looking for great experimentalists in various settings**